

ORIGINAL ARTICLE

Scheduling with present bias

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Abstract

The study of time-inconsistent decision-making, in particular present bias, has greatly influenced the field of behavioral economics over the last two decades. Moreover, it has resulted in the design and implementation of widely used incentive schemes for various important applications. However, the effect of present bias on decision-making for operational problems has been studied less. This work considers the effect of present bias in a simple scheduling system that requires decisions about project timing and sequencing. We design algorithms that enable optimization of revenue less cost under present bias for both naive and sophisticated (i.e., self-aware) decision-makers. We describe managerial insights about the relative performance of time-consistent, naive, and sophisticated decision-makers, and how to mitigate the effects of present bias. Both theoretical and computational results support these insights.

KEYWORDS

present bias, scheduling of projects, optimal timing, optimal sequencing

1 | INTRODUCTION

The study of *intertemporal preferences* in decision-making originates from the work of Phelps and Pollak (1968). Under intertemporal preferences, a decision-maker views the same trade-off differently at different times. Akerlof (1991) uses this concept to explain recurring procrastination as a result of greater salience for present costs or rewards than for future ones, that is, *present bias*. He discusses applications to purchasing, savings, gang behavior, and politics. Laibson (1994) presents a model that unifies conventional time-consistent discounting and present bias. As discussed in Section 2, present bias has been the subject of several empirical studies within behavioral economics.

O'Donoghue and Rabin (1999a) distinguish *naive* decision-makers who make decisions under the incorrect assumption that their future decisions will be time-consistent from *sophisticated* decision-makers who are aware of their future bias. In the context of a principal–agent model with asymmetric information, O'Donoghue and Rabin (1999b) design incentives for time-inconsistent procrastinating

agents. O'Donoghue and Rabin (2001) study a multiperiod model where a decision-maker chooses a unit-length task to process. The authors show that, due to present bias, an attractive new option can cause a decision-maker to switch from doing something beneficial to doing nothing. They further show that a decision-maker may procrastinate more to pursue important goals than unimportant ones. Thaler and Benartzi (2004) propose an employer-supported savings plan, *Save More TomorrowTM*, that encourages employee savings by mitigating present bias. According to Benartzi (2021), this plan has been used by more than 15 million employees.

The study of time-inconsistent preferences has received limited but increasing attention within the operations management literature. Su (2009) considers a problem where consumers delay purchases in situations where an immediate purchase is a rational choice. Such a delay is attributed to present bias in a situation when payment is immediate, and consumption is delayed. Plambeck and Wang (2013) compare charging options for a service with immediate costs and conclude that subscription charging is optimal. Gao et al. (2014) describe an optimal dynamic pricing policy for a manufacturer selling to customers with time-inconsistent preferences. Li et al. (2017) consider an environment that evolves

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stochastically, for which an optimal stopping policy is needed. A naive decision-maker follows a threshold stopping policy, and a sophisticated agent also does so if revenues are immediate. Kouvelis et al. (2018) address time inconsistency in a joint inventory and financial hedging problem in the mean-variance framework, where in each period a firm perceives a greater variability of the terminal wealth than the anticipated variability in a future period. Liao and Chen (2021) design a long-term conditional cash transfer program to encourage healthy habits assuming people's present-biased preferences. Li and Jiang (2022) study the influence of present bias on the pricing strategies of two competing firms that sell quality-differentiated products. Shi et al. (2022) demonstrate computationally that a considerable amount of project delay can be explained by present bias and design an incentive scheme that mitigates it. Zhang et al. (2022) investigate consumers' present bias in digital content consumption and their strategic self-control manners. In contrast with the above works, we consider decisions under present bias over both sequencing and timing decisions.

Closest to our application is the work of Wu et al. (2014), who consider a two-period model of a simple project without task precedence structure and with either a single worker or a team of workers. The effect of present bias is to reduce the work rate in the first period, which backloads work into the second period. A possible solution is to increase rewards for work that is completed in the first period. Our work differs from Wu et al. (2014) in two ways. First, we consider the effect of present bias on decisions that affect individual projects rather than an aggregate work rate. Second, we distinguish between decisions made by naive time-inconsistent decision-makers and those made by their sophisticated counterparts.

More generally, our study of present bias differs from all the above works in several ways. First, we consider both timing and sequencing decisions over multiple projects. Second, because we consider a sequence of many projects, we model the effect of present bias as it evolves sequentially based on previous decisions. Third, we examine trade-offs that arise between revenues and costs when the timing of both is altered by present bias. Fourth, we allow the level of present bias in decision-making to change over time.

There are many heuristic algorithms designed for the scheduling of projects, but these algorithms are primarily designed to alleviate the issue of computational intractability. By contrast, present bias exists for decisions over many problems, including those which are not computationally intractable. We observe that the number of projects to process is typically small for the behavioral decision-making environments studied in the literature. Our objective in this work is not to investigate optimization and approximation in the scheduling of projects from an algorithmic perspective, since there are many published works on this topic. Rather, our focus is on a more general question: how present bias, as modeled by a present bias coefficient and a level of self-awareness of the bias, affects decisions in scheduling of projects and how to mitigate biases in such decisions.

We summarize the contributions of our work. We introduce present bias into a simple scheduling system that requires project timing and sequencing decisions by a project company. We consider situations where cost is immediate and revenue occurs at project completion and the opposite case. For both timing and sequencing decisions, we design algorithms that enable optimization of revenue less cost under present bias for naive and sophisticated decision-makers. Based on this analysis, we describe managerial insights about the relative performance of time-consistent, naive, and sophisticated decision-makers, and how to mitigate the effects of present bias. Both theoretical and computational results support these insights. Applications of these insights are broad from understanding the efficiency of individual projects to influencing the design of project portfolios by managers.

This paper is organized as follows. Section 2 provides our notation and develops and contrasts time-consistent and time-inconsistent models for minimizing project scheduling costs. Section 3 studies the effect of present bias on timing decisions in scheduling. Sequencing decisions are similarly studied in Section 4. We provide a summary, general insights, and suggestions for future work in Section 5.

2 | NOTATION AND MODELS

Consider a project management company that schedules projects $N = \{1, \dots, n\}$ for different owners. The projects all require a scarce resource from the company and thus cannot be conducted concurrently. We study two scenarios for the company's decision. In the first scenario, the decision-maker faces a *timing* problem, where the projects follow a fixed sequence and must be processed in a given time window. In the second scenario, the decision-maker faces a *sequencing* problem, where projects must be sequenced and then processed sequentially. Idle time is allowed between projects in the timing problem but not in the sequencing problem. The decision-maker's objective is to maximize net profit, that is, revenue less cost. Note that a project can be rejected in both scenarios if it does not contribute to an overall greater profit to the decision-maker (Engels et al., 2003; Shabtay et al., 2013). In practice, a project can be contractually required or optional, and our work can model either assumption. By default, we assume that a project is optional. The decision problem includes either *immediate costs* or *immediate revenues*. With immediate costs, the decision-maker incurs a cost at the starting time of each project and receives revenue at its completion time. Conversely, with immediate revenues, the decision-maker receives a revenue at the starting time of each project and incurs a cost at its completion time.

A schedule σ is defined by the starting time $S_j(\sigma)$ and completion time $C_j(\sigma)$ of each project $j \in N$. Where the schedule is clear from context, we abbreviate $S_j(\sigma)$ and $C_j(\sigma)$ to S_j and C_j , respectively. Each completed project j earns the decision-maker a known revenue v_j and a cost f_j . As is typical in the

project management and scheduling literature, we assume the revenue v_j and cost f_j are functions of the project completion time C_j of project j . We allow revenue functions $v_1(\cdot), \dots, v_n(\cdot)$ and cost functions $f_1(\cdot), \dots, f_n(\cdot)$ to be defined arbitrarily with C_j . Each project has a positive processing time p_j and may also have an importance weight w_j and a due date d_j for certain cost functions. We assume that p_j and d_j are integer-valued, and once a project begins processing, no interruption or preemption is allowed until it is completed.

Since we consider decisions involving revenue and cost at different times, to capture the time value of money, we consider a discrete discount rate r . Evaluated at time 0, a revenue v earned at time t has a present value of $(1+r)^{-t}v$, and a cost f incurred at time t has a present value of $-(1+r)^{-t}f$.

We now formally define present bias in our timing and sequencing decisions. To model the decision-maker's present bias level, we introduce a present bias coefficient β_t at time t , $0 < \beta_t \leq 1$. Specifically, any value v occurred at a time later than t is worth only $\beta_t v$ at time t to the decision-maker. Note that t is the time at which a decision is made, as opposed to the timing of a cash flow. At any time t , if $\beta_t = 1$, then the decision-maker is time-consistent and has no present bias; if $0 < \beta_t < 1$, then the decision-maker views immediate revenues or costs as having a more significant relative value, and any future revenues or costs are weighted by β_t . We note that the present bias coefficient is fundamentally different from the discount rate. Present bias does not distinguish between an immediate slight delay and an immediate more considerable delay in the way that a discount factor does; it only distinguishes the existence of a delay. Under certain assumptions, present bias coefficients can model uncertainty of future values. In such cases, when the present bias coefficient increases over time, a higher level of uncertainty in the distant future is captured.

The present bias coefficient β , without distinguishing its time of occurrence and hence with subscript t omitted, has been subject to extensive empirical research within the behavioral economics literature. Existing works provide several empirical estimates of β , for example, 0.40 (Paserman, 2008), 0.78 (Meier & Sprenger, 2015), 0.82 (Kaur et al., 2015), 0.61 (Ericson, 2017), and 0.83 (Augenblick & Rabin, 2019). Examining 220 estimates of the present-bias parameter from 28 papers, Imai et al. (2021) find that those estimates vary considerably across studies, due primarily to the different characteristics of their applications.

With present bias and immediate costs, for a given schedule σ , at time zero, the decision-maker's intertemporal profit is

$$U_c^0(\sigma) = - \sum_{S_j(\sigma)=0} f_j(C_j(\sigma)) + \beta_0 \left[\sum_{j=1}^n (1+r)^{-C_j(\sigma)} v_j(C_j(\sigma)) - \sum_{S_j(\sigma)>0} (1+r)^{-S_j(\sigma)} f_j(C_j(\sigma)) \right]. \quad (1)$$

Alternatively, with present bias and immediate revenues, for a given schedule σ , the decision-maker's intertemporal

profit at time zero is

$$U_r^0(\sigma) = \sum_{S_j(\sigma)=0} v_j(C_j(\sigma)) + \beta_0 \left[\sum_{S_j(\sigma)>0} (1+r)^{-S_j(\sigma)} v_j(C_j(\sigma)) - \sum_{j=1}^n (1+r)^{-C_j(\sigma)} f_j(C_j(\sigma)) \right]. \quad (2)$$

In both (1) and (2), parameter β_0 , $0 < \beta_0 \leq 1$, specifies the level of present bias of the decision-maker at time zero. This definition applies at any time $t > 0$, for an integer value of t and a present bias coefficient β_t at time t . We adopt a commonly used rate to model the degree of present bias. Other forms of the present bias discount include fixed costs (Benhabib et al., 2010).

A *time-inconsistent decision-maker* with $0 < \beta_t < 1$ at time t is of one of two types. A *sophisticated decision-maker* (*sophisticate*) knows that he has present bias and assumes that he will have present bias in future decisions; while a *naïve decision-maker* (*naïf*) applies present bias in his immediate decision but incorrectly assumes he will be time consistent in the future. Our modeling of the decision biases of the naïf and sophisticate follows the behavioral economics literature (e.g., O'Donoghue & Rabin, 1999a).

For a naïf, even knowing time-consistent decisions does not necessarily lead to implementing such decisions. At period t , decisions are determined that should be implemented at periods $t, t+1, \dots$. In planning those decisions, a naïf assumes that he will make time-consistent decisions at periods $t+1, \dots$. However, once period $t+1$ arrives, the actual decision made by the naïf is still affected by present bias. Thus, the real preference in the future is different from the naïf's believed future preference, and this results in biased decision-making. On the other hand, a sophisticate has the correct beliefs about his future preferences. In making decisions at time t , a sophisticate correctly assumes that, at periods $t+1, \dots$, he will make decisions under present bias. Subsequently, he will actually follow those decisions. The concept of decision-makers' awareness of their present bias can be extended to situations between naivety and sophistication, namely *partial naivete* (O'Donoghue & Rabin, 2001).

As mentioned above, the concepts of naivety and sophistication used in our work closely follow the literature of behavioral economics; however, they can also be interpreted in a more general context. The naïve decision-maker is naïve only about the present bias that affects their decision-making, and their cognitive ability is not affected. Therefore, it is reasonable for them to assume that they will make a fully rational decision in the future. Such behavior is commonly observed in situations of repeated procrastination, such as, "I will quit smoking tomorrow" or "I will start saving for retirement next year" (O'Donoghue & Rabin, 2015). Observe that, in such situations, the procrastinating behavior does not reduce the decision-maker's understanding of what constitutes positive behavior.

On the other hand, the sophisticate's awareness of their present bias does not enable them to correct it. O'Donoghue

and Rabin (2002) describe their solutions as playing a game against their future selves. Thus, their behavior partly reflects negative behavior by their future selves, which they cannot control, and partly reflects attempts to induce positive behavior from their future selves. An example would be delaying the opening of a food or drink item due to an awareness of one's tendency to finish it immediately once opened. Various practical situations related to addictions exemplify this. Observe that, in such situations, the decision-maker's awareness of their addiction does not by itself eliminate the addiction.

3 | TIMING DECISIONS

3.1 | Time-consistent decisions

We illustrate the timing decisions of a time-consistent decision-maker (TC). We assume the planning horizon is from 0 to T . Consistent with the classical scheduling literature, the time horizon $[0, T]$ is discretized, where a decision can only be made at integer time points. For feasibility of scheduling all projects, we assume the total processing time of all projects, $P = \sum_{j=1}^n p_j$, satisfies $P \leq T$. We have a classical scheduling problem for time-consistent decision-making. Further, if the scheduling cost function f_j of each project is *regular*, that is, nondecreasing with project completion time C_j , then a TC schedules each project as early as possible without inserted idle time between projects. However, we show below that a time-inconsistent decision-maker may procrastinate.

For the case with immediate costs, the time-consistent decision-making problem can be solved by the following dynamic program.

Algorithm time consistent for timing (TCT)

Input

Given $T, r, p_1, \dots, p_n, f_1(\cdot), \dots, f_n(\cdot)$, and $v_1(\cdot), \dots, v_n(\cdot)$, where projects are to be processed in sequence $1, \dots, n$.

Value function

$H^{TC}(j, t)$ = the maximum total net profit of a TC from any schedule of projects $\{j, \dots, n\}$ in which the starting time of project j is no earlier than t .

Boundary condition

$$H^{TC}(j, t) = \begin{cases} 0, & \text{if } j = n + 1 \text{ and } 0 \leq t \leq T, \\ -\infty, & \text{if } 1 \leq j \leq n + 1 \text{ and } t > T. \end{cases} \quad (3)$$

Optimal solution value

$H^{TC}(1, 0)$.

Recurrence relation

$$H^{TC}(j, t) = \max \begin{cases} H^{TC}(j, t + 1), \\ (1 + r)^{-(t+p_j)} v_j - (1 + r)^{-t} f_j(t + p_j) \\ \quad + H^{TC}(j + 1, t + p_j), \\ H^{TC}(j + 1, t), \end{cases} \quad (4)$$

where the first alternative schedules idle time from t to $t + 1$, the second alternative schedules project j with starting time t and completion time $t + p_j$ and hence revenue v_j is received and cost $f_j(t + p_j)$ is incurred, and the third alternative rejects project j .

Also, for the case with immediate revenues, the time-consistent decision-making problem can be solved by a similar algorithm. The only change is that the second alternative in the recurrence relation becomes $(1 + r)^{-t} v_j - (1 + r)^{-(t+p_j)} f_j(t + p_j) + H^{TC}(j + 1, t + p_j)$.

3.2 | Time-inconsistent decisions

We consider time-inconsistent decisions with immediate costs and revenues at project completion in Section 3.2.1 and with immediate revenues and costs at project completion in Section 3.2.2.

3.2.1 | Immediate costs

We consider a naïf or sophisticate's timing decisions for multiple projects with immediate costs. Recall that a naïf assumes in the future he will make time-consistent decisions. The value of a naïf's timing decision is defined by the function $H_c^N(j, t)$, where $H^{TC}(\cdot)$ is defined in Algorithm TCT.

Algorithm naïf immediate costs for timing (NICT)

Value function

$H_c^N(j, t)$ = the maximum total net profit perceived by a naïf at time t of any schedule of projects $\{j, \dots, n\}$, where the starting time of project j is no earlier than t .

Boundary condition

$$H_c^N(j, t) = \begin{cases} 0, & \text{if } j = n + 1 \text{ and } 0 \leq t \leq T, \\ -\infty, & \text{if } 1 \leq j \leq n + 1 \text{ and } t > T. \end{cases} \quad (5)$$

Recurrence relation

$$H_c^N(j, t) = \max \begin{cases} \beta_t H^{TC}(j, t + 1), \\ -(1 + r)^{-t} f_j(t + p_j) \\ \quad + \beta_t [(1 + r)^{-(t+p_j)} v_j(t + p_j) \\ \quad + H^{TC}(j + 1, t + p_j)], \\ H_c^N(j + 1, t), \end{cases} \quad (6)$$

where, in the first alternative, the decision-maker procrastinates by scheduling idle time from t to $t + 1$ and assumes that he will make a time-consistent decision at time $t + 1$ for projects $j, \dots, n$. In the second alternative, the decision-maker schedules project j to start at time t and completes at time $t + p_j$ with an immediate cost of $(1 + r)^{-t} f_j(t + p_j)$ and a perceived revenue of $\beta_t (1 + r)^{-(t+p_j)} v_j(t + p_j)$ and assumes that he will make a time-consistent decision at time

TABLE 1 TC's values in Example 1

$j \backslash t$	0	1	2	3	4	5	6
1	6.5	4.5	3	2	1	0	$-\infty$
2	5	4	3	2	1	0	$-\infty$
3	0	0	0	0	0	0	$-\infty$

TABLE 2 Naif's perceived values in Example 1

$j \backslash t$	0	1	2	3	4	5	6
1	1.8	1.2	0.8	0.4	0	0	$-\infty$
2	1.6	1.2	0.8	0.4	0	0	$-\infty$
3	0	0	0	0	0	0	$-\infty$

$t + p_j$ for projects $j + 1, \dots, n$. However, the naif will actually make a decision that is again subject to present bias at any future time. In the third alternative, the decision-maker rejects project j at time t and continues scheduling projects $j + 1, \dots, n$ from a time no earlier than t with present bias.

Due to present bias in decision-making, a naif may procrastinate project processing. Consider the following example.

Example 1. The time horizon is $T = 5$, and the discount rate is $r = 0$. The scheduling cost equals project completion time, that is, $f_j(C_j) = C_j$, and the present bias coefficient is $\beta = 0.4$ at any time. There are two projects with processing times $p_1 = 2$ and $p_2 = 1$ and constant values $v_1 = 5.5$ and $v_2 = 6$.

For Example 1, the TC's profits characterized by function $H^{TC}(j, t)$ for different values of j and t are provided in Table 1. Optimally, the TC processes projects 1 and 2 in the time intervals $[0, 2]$ and $[2, 3]$, respectively, with a total profit of $(5.5 - 2) + (6 - 3) = 6.5$. The optimal solution path is highlighted in Table 1.

The naif's profits defined by function $H_c^N(j, t)$ for different values of j and t in Example 1 are shown in Table 2. To see how the values in Table 2 are obtained, consider the value for $j = 1$ and $t = 0$, that is, $H_c^N(1, 0)$. At time 0, if the naif procrastinates by scheduling idle time from 0 to 1 and assumes that he will make a time-consistent decision at time 1 for projects 1 and 2, then his perceived profit is $\beta H^{TC}(1, 1) = 0.4 * 4.5 = 1.8$. If he processes project 1 in the time interval $[0, 2]$ and assumes he will make a time-consistent decision at time 2 for project 2, then his perceived profit is $-f_1(2) + \beta[v_1(2) + H^{TC}(2, 2)] = -2 + 0.4(5.5 + 3) = 1.4$. If he rejects project 1 at time 0 and continues scheduling project 2 from time 0 with present bias, then his perceived profit is $H_c^N(2, 0)$. Regarding $H_c^N(2, 0)$, if the naif procrastinates by scheduling idle time from 0 to 1 and assumes that he will make a time-consistent decision at time 1 for project 2, then his perceived profit is $\beta H^{TC}(2, 1) = 0.4 * 4 = 1.6$. If he processes project 2 in the time interval $[0, 1]$, then his per-

TABLE 3 Sophisticate's values in Example 1

$j \backslash t$	0	1	2	3	4	5	6
1	1.4	0.4	0	0	0	0	$-\infty$
2	1.4	0.4	0	0	0	0	$-\infty$
3	0	0	0	0	0	0	$-\infty$

ceived profit is $-f_2(1) + \beta v_2(1) = -1 + 0.4 * 6 = 1.4$. If he rejects project 2 at time 0, then his perceived profit is 0. Therefore, we have that $H_c^N(2, 0) = \max\{1.6, 1.4, 0\} = 1.6$ and $H_c^N(1, 0) = \max\{1.8, 1.4, 1.6\} = 1.8$. This implies that the naif waits until time 1 to make the next decision, which is characterized by value function $H_c^N(1, 1)$. Following a similar analysis as above, we find that at time 1, the naif will again not process project 1. Continuing the decision process, the naif ends up doing nothing and earns an actual profit of 0, though at time 0 he perceives a profit of 1.8. One optimal solution path of the naif is highlighted in Table 2.

We next consider a sophisticate.

Algorithm sophisticated immediate costs for timing (SICT)

Value function

$H_c^S(j, t)$ = the maximum total net profit perceived by a sophisticate at time t from any schedule of projects $\{j, \dots, n\}$ where the starting time of project j is no earlier than t .

Boundary condition

$$H_c^S(j, t) = \begin{cases} 0, & \text{if } j = n + 1 \text{ and } 0 \leq t \leq T, \\ -\infty, & \text{if } 1 \leq j \leq n + 1 \text{ and } t > T. \end{cases} \quad (7)$$

Recurrence relation

$$H_c^S(j, t) = \max \begin{cases} \beta_t H_c^S(j, t + 1), \\ -(1 + r)^{-t} f_j(t + p_j) \\ + \beta_t [(1 + r)^{-(t+p_j)} v_j(t + p_j) \\ + H_c^S(j + 1, t + p_j)], \\ H_c^S(j + 1, t). \end{cases} \quad (8)$$

Remark 1. For a TC, the perceived profit is equal to the realized profit if he follows Algorithm TCT's decisions. For a sophisticate, the scheduling decisions he plans are the same as he later implements. The true net profit from the schedule, as adjusted by present bias, equals the sophisticate's perceived profit when he makes his scheduling decisions. However, this is not the case for a naif because a naif's anticipated future decisions may not match the actual decisions he will make in the future.

The sophisticate's profits defined by function $H_c^S(j, t)$ for different values of j and t are shown in Table 3. Specifically, the sophisticate rejects project 1 at time 0, and processes

project 2 in the time interval $[0,1]$, resulting in an intertemporal profit of $-1 + 0.4 * 6 = 1.4$, though the true profit he earns is $-1 + 6 = 5$. The sophisticate's optimal solution path is highlighted in Table 3.

From Example 1, procrastination of project processing has different impacts on the naif's and sophisticate's intertemporal profits. This example illustrates how, with immediate costs, regular scheduling costs, and reasonably large project values, a naif procrastinates more than a sophisticate. We have the following related result.

Proposition 1. *Consider projects with immediate costs, namely projects $j, \dots, n$, to be processed in that sequence with starting time no earlier than time t . Assume $r = 0$; and the present bias parameter and project values are time independent.*

If project j is not rejected, and

$$f_j(t + p_j) \leq \beta f_j(t + p_j + k), \quad \forall k = 1, \dots, T - \sum_{i=j}^n p_i - t, \quad (9)$$

then a naif starts to process project j at time t to maximize his intertemporal profit.

If project j is not rejected, and

$$f_j(t + p_j) \leq \beta^k f_j(t + p_j + k) + \beta(1 - \beta^k)v_j, \quad \forall k = 1, \dots, T - \sum_{i=j}^n p_i - t, \quad (10)$$

then a sophisticate starts to process project j at time t to maximize his intertemporal profit.

Proof. Note that project j can only be feasibly processed with starting time $t, t + 1, \dots, T - \sum_{i=j}^n p_i - t$. First, consider a naif's decision. Suppose a naif is idle from t to $t + k$ and processes project j from $t + k$ to $t + p_j + k$, for an integer value of k between 1 and $T - \sum_{i=j}^n p_i - t$. The intertemporal profit of project j for the naif at time t is $-\beta f_j(t + p_j + k) + \beta v_j$. This profit is consistent with the second alternative of function $H_c^N(j, t)$ in Algorithm NICT, where the naif is idle from time t to $t + 1$, and the present bias coefficient β applies to the profit and cost of project j incurred no earlier than time $t + 1$. Note that the naif assumes his decisions from time $t + 1$ are time consistent, and so no further present bias applies at time $t + 1$ or later. If the naif processes project j from time t to $t + p_j$ instead, then his intertemporal profit of project j at time t becomes $-f_j(t + p_j) + \beta v_j$, as in the second alternative for project j of function $H_c^N(j, t)$ in Algorithm NICT with $r = 0$. The processing time of every other project is unchanged. Thus, under condition (9), the naif's perceived profit is at least as large when processing project j from time t .

Next, consider a sophisticate's decision. Suppose a sophisticate is idle from t to $t + k$ and processes project j from

$t + k$ to $t + p_j + k$, for an integer value of k between 1 and $T - \sum_{i=j}^n p_i - t$. The intertemporal profit of project j for the sophisticate at time t is $-\beta^k f_j(t + p_j + k) + \beta^{k+1} v_j$, as defined by the second alternative of function $H_c^S(j, t)$ for k recurrences in Algorithm SICT. Note that the first alternative of function $H_c^S(j, t)$ applies the present bias coefficient for each unit length of the idle time, and thus when there is an idle time of length k , with an immediate cost, compound coefficients β^k and β^{k+1} apply to the project cost and value, respectively. If the sophisticate processes project j from time t to $t + p_j$ instead, then his intertemporal profit of project j at time t becomes $-f_j(t + p_j) + \beta v_j$, as in the first alternative for project j of function $H_c^S(j, t)$ in Algorithm SICT. The processing time of every other project is unchanged. Thus, under condition (10), the sophisticate's profit is at least as much when processing project j starting at time t instead of $t + k$. $\square$

For a TC, a regular scheduling cost is sufficient to prevent any procrastination of project processing. For a naif, to eliminate procrastination, a stronger condition on scheduling cost is needed, as specified by condition (9); that is, the present scheduling cost needs to be no more than the future scheduling cost adjusted by the present bias coefficient. For a sophisticate, project revenues play a role in scheduling decisions, as characterized by condition (10). Intuitively, when a project's revenue is sufficiently large relative to its cost, it is not procrastinated by a sophisticate.

3.2.2 | Immediate revenues

Algorithm naive immediate revenues for timing (NIRT)

Value function

$H_r^N(j, t)$ = the maximum total net profit perceived by a naif at time t for any schedule of projects $\{j, \dots, n\}$ in which the starting time of project j is no earlier than t .

Boundary condition

$$H_r^N(j, t) = \begin{cases} 0, & \text{if } j = n + 1 \text{ and } 0 \leq t \leq T, \\ -\infty, & \text{if } 1 \leq j \leq n + 1 \text{ and } t > T. \end{cases} \quad (11)$$

Recurrence relation

$$H_r^N(j, t) = \max \begin{cases} \beta_t H^{TC}(j, t + 1), \\ (1 + r)^{-t} v_j(t + p_j) \\ + \beta_t [-(1 + r)^{-(t+p_j)} f_j(t + p_j) \\ + H^{TC}(j + 1, t + p_j)], \\ H_r^N(j + 1, t). \end{cases} \quad (12)$$

To show how present bias can affect a decision-maker's choice, consider the following example where the scheduling cost is linearly decreasing in the project completion time. Such a cost structure applies when projects have specific due

TABLE 4 TC's values in Example 2

$j \backslash t$	0	1	2	3	4	5	6
1	3.5	3.5	3.5	3	2.5	0	$-\infty$
2	3	3	3	3	0	0	$-\infty$
3	0	0	0	0	0	0	$-\infty$

TABLE 5 Naif's perceived values in Example 2

$j \backslash t$	0	1	2	3	4	5	6
1	4.56	4.8	5.04	5.28	4.78	0	$-\infty$
2	4.56	4.8	5.04	5.28	0	0	$-\infty$
3	0	0	0	0	0	0	$-\infty$

dates, and earliness costs occur when a project is completed before its due date (Garey et al., 1988).

Example 2. The scheduling cost is $f_j(C_j) = 8 - C_j$, and the present bias coefficient is $\beta = 0.24$ at any time. There are two projects with processing times $p_1 = 1$ and $p_2 = 2$ and fixed revenues $v_1 = 5.5$ and $v_2 = 6$. The time horizon is $T = 5$, and the discount rate is $r = 0$.

For Example 2, the TC's profits defined by function $H^{TC}(j, t)$ are shown in Table 4. The TC strategically postpones the processing of his two projects to be in the time intervals of $[2, 3]$ and $[3, 5]$, resulting in a total profit of $[5.5 - (8 - 3)] + [6 - (8 - 5)] = 3.5$. The optimal solution path is highlighted in Table 4.

The naif's profits defined by function $H_r^N(j, t)$ in Example 2 are shown in Table 5. To see how the values in Table 5 are obtained, consider the value for $j = 1$ and $t = 0$, that is, $H_r^N(1, 0)$. At time 0, if the naif procrastinates by scheduling idle time from 0 to 1 and assumes that he will make a time-consistent decision at time 1 for projects 1 and 2, then his perceived profit is $\beta H^{TC}(1, 1) = 0.24 * 3.5 = 0.84$; if he processes project 1 in the time interval $[0, 1]$ and assumes he will make a time-consistent decision at time 1 for project 2, then his perceived profit is $v_1(1) + \beta(-f_1(1) + H^{TC}(2, 1)) = 5.5 + 0.24(-(8 - 1) + 3) = 4.54$; if he rejects project 1 at time 0 and continues scheduling project 2 from time 0 with present bias, then his perceived profit is $H_r^N(2, 0)$. Regarding $H_r^N(2, 0)$, if the naif procrastinates by scheduling idle time from 0 to 1 and assumes that he will make a time-consistent decision at time 1 for project 2, then his perceived profit is $\beta H^{TC}(2, 1) = 0.24 * 3 = 0.72$; if he processes project 2 in the time interval $[0, 2]$, then his perceived profit is $v_2(2) - \beta f_2(2) = 6 - 0.24 * (8 - 2) = 4.56$; if he rejects project 2 at time 0, then his perceived profit is 0. Therefore, we have that $H_r^N(2, 0) = \max\{0.72, 4.56, 0\} = 4.56$ and $H_r^N(1, 0) = \max\{0.84, 4.54, 4.56\} = 4.56$. Thus, the naif rejects project 1 at time 0 and processes project 2 from time 0 to 2, perceiving

TABLE 6 Sophisticate's values in Example 2

$j \backslash t$	0	1	2	3	4	5	6
1	4.972	5.2696	5.5672	5.28	4.78	0	$-\infty$
2	4.56	4.8	5.04	5.28	0	0	$-\infty$
3	0	0	0	0	0	0	$-\infty$

a profit of 4.56 with present bias. However, the true profit he earns is $6 - (8 - 2) = 0$. The naif's optimal solution path is highlighted in Table 5.

Similarly, a sophisticate's timing decision over multiple projects on a single facility can be characterized by the function $H_r^S(j, t)$ with the following recurrence relation.

Algorithm sophisticated immediate revenues for timing (SIRT)

Value function

$H_r^S(j, t)$ = the maximum total net profit perceived by a sophisticate at time t for any schedule of projects $\{j, \dots, n\}$ in which the starting time of project j is no earlier than t .

Boundary condition

$$H_r^S(j, t) = \begin{cases} 0, & \text{if } j = n + 1 \text{ and } 0 \leq t \leq T, \\ -\infty, & \text{if } 1 \leq j \leq n + 1 \text{ and } t > T. \end{cases} \quad (13)$$

Recurrence relation

$$H_r^S(j, t) = \max \begin{cases} \beta_t H_r^S(j, t + 1), \\ (1 + r)^{-t} v_j(t + p_j) \\ + \beta_t [-(1 + r)^{-(t+p_j)} f_j(t + p_j) \\ + H_r^S(j + 1, t + p_j)], \\ H_r^S(j + 1, t), \end{cases} \quad (14)$$

where the first alternative procrastinates by scheduling idle time from t to $t + 1$. The second alternative schedules project j to start at time t and completes at time $t + p_j$, and hence a revenue of $(1 + r)^{-t} v_j(t + p_j)$ is earned and a cost of $(1 + r)^{-(t+p_j)} f_j(t + p_j)$ is perceived with present bias. Except for the immediate revenue, all revenues and costs are adjusted by the present bias coefficient β_t . The third alternative rejects project j at time t and continues scheduling projects $j + 1, \dots, n$ from time t with present bias.

For Example 2, a sophisticate's profit defined by $H_r^S(j, t)$ is presented in Table 6. The sophisticate processes the two projects in the time intervals $[0, 1]$ and $[1, 3]$, with present bias, perceiving a total profit of $[5.5 - 0.24 * (8 - 1)] + 0.24 * [6 - 0.24 * (8 - 3)] = 4.972$. However, the true value without present bias he earns in this schedule is only $[5.5 - (8 - 1)] + [6 - (8 - 3)] = -0.5$. The sophisticate's optimal solution path is highlighted in Table 6.

Example 2 demonstrates that with immediate revenues, project processing can be preoperated by a decision-maker with present bias. We next characterize conditions under which procrastination of project processing is unnecessary with immediate revenues.

Proposition 2. *In a situation with immediate revenues, consider projects $j, \dots, n$, to be processed in that sequence with starting time no earlier than time t . Assume $r = 0$; and the present bias parameter and project values are time independent.*

If project j is not rejected, and

$$f_j(t + p_j) \leq f_j(t + p_j + k) + (1/\beta - 1)v_j, \quad \forall k = 1, \dots, T - \sum_{i=j}^n p_i - t, \quad (15)$$

then a naïf starts to process project j at time t to maximize his intertemporal profit.

If project j is not rejected, and

$$f_j(t + p_j) \leq \beta^k f_j(t + p_j + k) + (1/\beta - \beta^{k-1})v_j, \quad \forall k = 1, \dots, T - \sum_{i=j}^n p_i - t, \quad (16)$$

then a sophisticate starts to process project j at time t to maximize his intertemporal profit.

Proof. Project j can only feasibly start at times $t, t + 1, \dots, T - \sum_{i=j}^n p_i - t$. Consider a naïf's decision. Suppose a naïf processes a project j from $t + k$ to $t + p_j + k$, for an integer value of k between 1 and $T - \sum_{i=j}^n p_i - t$. The intertemporal profit of project j for the naïf at time t is $\beta v_j - \beta f_j(t + p_j + k)$. This profit is consistent with the first alternative of function $H_r^N(j, t)$ in Algorithm NIRT, where the naïf is idle from time t to $t + 1$, and the present bias coefficient β applies to the profit and cost of project j incurred no earlier than time $t + 1$. Since the naïf assumes his decisions from time $t + 1$ are time consistent, no further present bias applies at time $t + 1$ or later. If project j starts at t , then the naïf's intertemporal profit of project j at time t becomes $v_j - \beta f_j(t + p_j)$, as in the second alternative for project j of function $H_r^N(j, t)$ in Algorithm NIRT. The processing of every other project is unchanged. Thus, under condition (15), processing project j from t maximizes the naïf's perceived profit.

Now, consider a sophisticate's decision. Suppose a sophisticate starts a project j at time $t + k$, for an integer value of k between 1 and $T - \sum_{i=j}^n p_i - t$. The perceived profit of the sophisticate at time t is $\beta^k v_j - \beta^{k+1} f_j(t + p_j + k)$, as in the first alternative for project j of function $H_r^S(j, t)$ for k recurrences in Algorithm SIRT with $r = 0$. Note that the first alternative of function $H_r^S(j, t)$ applies the present bias coefficient for each

unit length of the idle time, and thus when there is an idle time in the length of k , with an immediate value, compound coefficients β^k and β^{k+1} apply to the project value and cost, respectively. When the sophisticate processes project j from time t , his intertemporal profit at time t is $v_j - \beta f_j(t + p_j)$, as in the second alternative for project j of function $H_r^S(j, t)$ in Algorithm SIRT with $r = 0$. The processing time of every other project is unchanged. Thus, under condition (16), the sophisticate's profit is maximized when processing project j from t . $\square$

Remark 2. Observe that condition (15) is weaker than regularity of scheduling cost. Therefore, with regular scheduling costs and without rejection, a naïf's timing decision is the same as that of a TC, where no idle time occurs between processing any two projects. However, with general scheduling costs, a naïf may process projects too early due to present bias. For a sophisticate, condition (16) is weaker than condition (10) in Proposition 1. Therefore, for a sophisticate, a smaller project revenue can lead to earlier processing of a project with immediate revenues compared to the situation with immediate costs.

3.3 | Managerial insights for timing decisions

We investigate how present bias affects a naïf and a sophisticate's timing decision with scheduling cost defined as $f_j(C_j) = C_j$. This cost structure represents a situation where the projects are of similar importance; hence, their costs are well modeled by their completion times. We present our main results as follows.

Theorem 1. *If $r = 0$, $f_j(C_j) = C_j$, and the present bias parameter is time independent,*

- (i) *with immediate costs, either a naïf or a sophisticate may procrastinate project processing;*
- (ii) *with immediate revenues, a naïf processes his projects without inserted idle time; but a sophisticate may procrastinate. However, if v_j is independent of project completion time and $v_j \geq \sum_{i=1}^j p_i$, then a sophisticate does not procrastinate.*

Proof. Part (i) follows from Example 1. For part (ii), with immediate revenues, a naïf will not procrastinate, as shown by Remark 2. For a sophisticate, consider an example with present bias coefficient $\beta = 0.4$, one project with processing time 1 and value 1, and $T = 2$. The sophisticate perceives a profit of $1 - 0.4 * 1 = 0.6$ for processing the project from 0 to 1, compared to a profit of $1 - 0.4 * (0.4 * 2) = 0.68$ for processing it from 1 to 2, and hence will procrastinate.

Now, consider the case where $v_j \geq \sum_{i=1}^j p_i$ and projects $1, \dots, j$ are all processed. By induction, assume projects 1 through $j - 1$ are processed without inserted idle time. We show that inequality (16) holds when $t = \sum_{i=1}^{j-1} p_i$. On the

right side of inequality (16), we have

$$\begin{aligned}
 & \beta^k f_j \left(\sum_{i=1}^j p_i + k \right) + (1/\beta - \beta^{k-1}) v_j \\
 & \geq \beta^k \left(\sum_{i=1}^j p_i + k \right) + (1/\beta - \beta^{k-1}) \sum_{i=1}^j p_i \\
 & \geq [1/\beta - \beta^{k-1}(1 - \beta)] \sum_{i=1}^j p_i \\
 & \geq [1/\beta - (1 - \beta)] \sum_{i=1}^j p_i \\
 & \geq \sum_{i=1}^j p_i,
 \end{aligned} \tag{17}$$

where the last inequality follows from $1/\beta + \beta \geq 2$. Hence, inequality (16) holds and the sophisticate does not procrastinate.

If some projects among $1, \dots, j$ are rejected, we can remove the rejected projects and reindex the remaining ones, and the above induction argument still holds. Hence, the sophisticate does not procrastinate. $\square$

Theorem 1 shows that immediate revenues can help prevent procrastination for the timing problem. We observe that the cost function $f_j(C_j) = C_j$ maps completion time into cost with appropriate scaling. The condition $v_j \geq \sum_{i=1}^j p_i$ used in Theorem 1 can be interpreted more generally as $v_j \geq f_j(C_j)$. This inequality is reasonable in that it merely requires that a project is profitable when all projects before it are processed without idle time. We perform the following numerical experiment to see how immediate costs can lead to reduced profit due to present bias.

The number of projects is fixed at $n = 10$, which is sufficient to illustrate the effects of present bias in behavioral decisions, though computationally we can solve much larger instances. The processing time of each project $j \in N$ is an integer randomly generated from the uniform integer distribution $UI[1, 10]$. For project j , the value is randomly generated as $v_j = \sum_{i=1}^j p_i + \alpha v$, where α is a scaling parameter representing the magnitude of project values relative to project processing times, and v is randomly generated from the uniform integer distribution $UI[1, 10]$. This design follows a principle of Hall and Posner (2001) to ensure that if $\alpha > 0$ the net profit of each individual project is positive. However, note that project values generated in this way are insufficient to prevent project rejection by a TC, naif, or sophisticate. Let $t = T - P$ be the extra time available for the timing decisions. For the four parameters we control, we consider $r = 0$, $\beta \in \{0.4, 0.45, \dots, 0.95\}$, $\alpha \in \{1.5, 1.55, \dots, 2.5\}$, and $t \in \{1, 2, \dots, 20\}$. The outcome of interest is the profit loss percentage compared to a TC at time 0. We randomly gen-

erate 30 instances for each parameter setting and report the average results.

We observe that the profit loss from present bias can result from both procrastination and overrejection. To separate these two effects, we consider procrastination and rejection separately, by first assuming that all projects are contractually required. In the following, each figure is based on results from all the instances. That is, for the value of one parameter under study (e.g., β in Figure 1), we use the average from instances with all the values of other parameters (e.g., α and t for Figure 1). First, we require all the projects to be processed and obtain results in Figures 1–3. Figure 1 shows that, as expected, a higher degree of present bias represented by smaller values of β causes a more significant loss of profit. Also, a naif incurs greater profit loss than a sophisticate; interestingly, this difference increases with β when β is relatively small but decreases with β when β is relatively large.

Figure 2 demonstrates that the profit loss decreases with the ratio of project values to project processing times. Note that, in our instances, larger processing times lead to greater scheduling costs. The pattern in Figure 2 occurs because the timing decision does not impact project values, and the relative profit loss decreases as the projects become more valuable. Figure 3 shows that with a longer time horizon the loss of profit from present bias is more significant. This echoes the following well-known phenomenon in project management: a looser deadline causes more project delays (Gutierrez & Kouvelis, 1991).

Next, we assume that, in the absence of contractual obligations, the project company can choose which projects to process. To focus on the rejection issue, we assume that project processing starts from time 0, and there is no inserted idle time between any two projects. Thus, present bias causes no procrastination here. For our instances, on average, a TC rejects 3.60 out of the 10 available projects. A naif and sophisticate reject 7.58 and 6.17 projects and incur profit losses of 44.3% and 20.0%, respectively. Intuitively, a naif overestimates his rationality in the future and consequently overrates his future profit. As a result, he is more willing to forego a current project to reduce the cost of future projects; however, those future projects will not be processed as he expects, and additional profit loss will occur. On the other hand, a sophisticate has a better understanding of his present bias in the future and incurs a less severe overrejection. The profit loss due to overrejection is shown in Figures 4 and 5. Figure 4 shows that profit loss due to overrejection decreases where there is a low level of present bias, that is, a high β value. At most levels of present bias, a naif's profit loss percentage approximately doubles that of a sophisticate. Figure 5 shows that the relative profit loss percentage caused by overrejection is insensitive to the relative value of projects compared to their costs.

We now discuss our managerial insights from the above analysis in Sections 3.1–3.3. We are interested in how far a naif or a sophisticate's decision deviates from that of a TC and what mechanisms we can use to mitigate the resulting loss of profit.

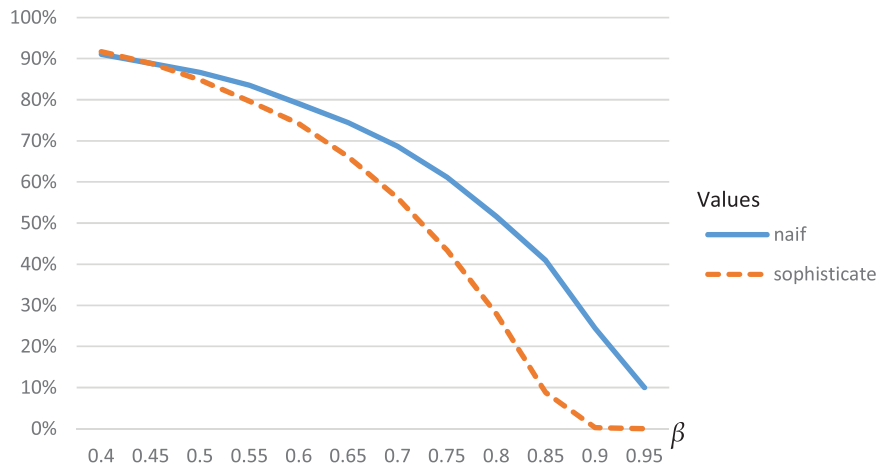


FIGURE 1 Percentage of profit loss by procrastination as a function of β [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1111/poms.13937)]

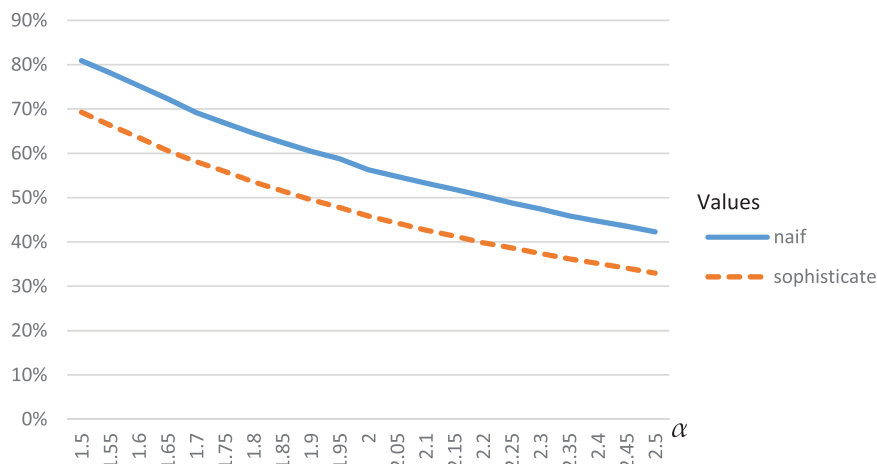


FIGURE 2 Percentage of profit loss by procrastination as a function of α [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1111/poms.13937)]

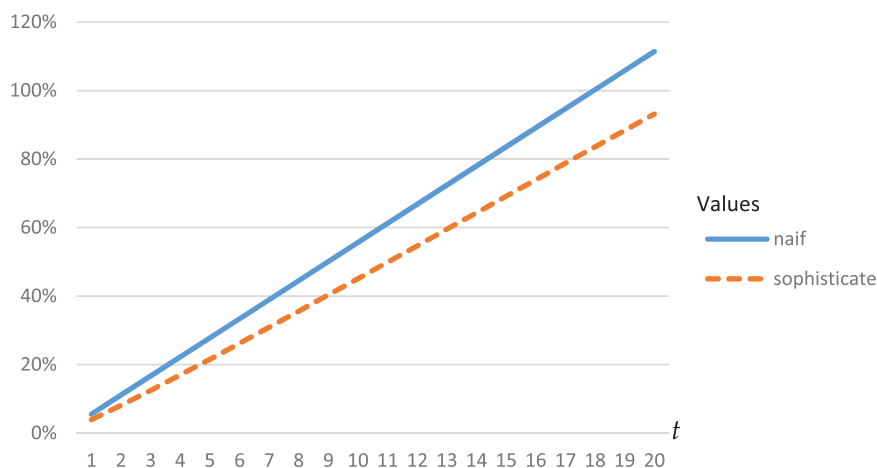


FIGURE 3 Percentage of profit loss by procrastination as a function of t [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1111/poms.13937)]

First, we find that, in general, immediate costs incentivize a naif or sophisticate to delay his project processing or even reject projects. This is because an immediate cost is overvalued relative to future values of projects by a present-biased decision-maker. In contrast, immediate revenues encourage a naif or sophisticate to advance his project processing more than optimally due to the discounted future cost perceived by a biased decision-maker. Therefore, under present bias, to

avoid delay or rejection of an urgent project, a project owner may offer a contract with immediate revenues; and to postpone a project strategically, a project owner may offer one with immediate costs.

Second, when project revenues are sufficient to cover their costs, with either immediate revenues or costs, a naif may procrastinate project processing or reject projects more than a sophisticate. On the other hand, a sophisticate may

FIGURE 4 Percentage of profit loss by rejection as a function of β [Color figure can be viewed at wileyonlinelibrary.com]

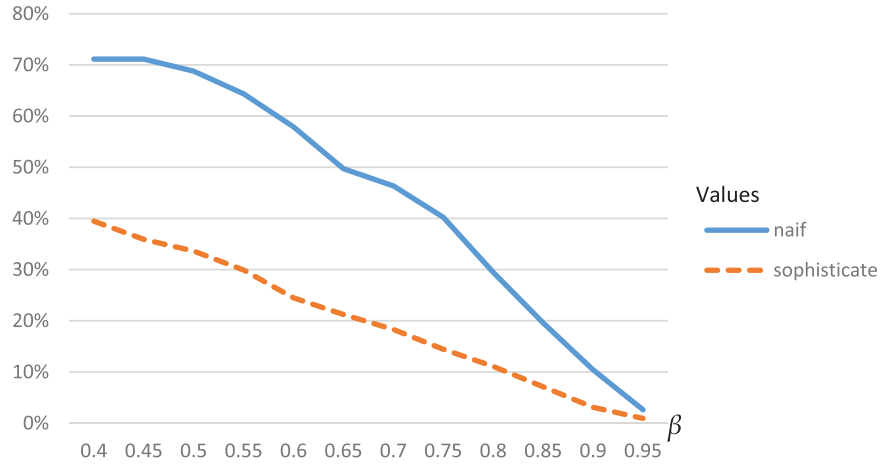
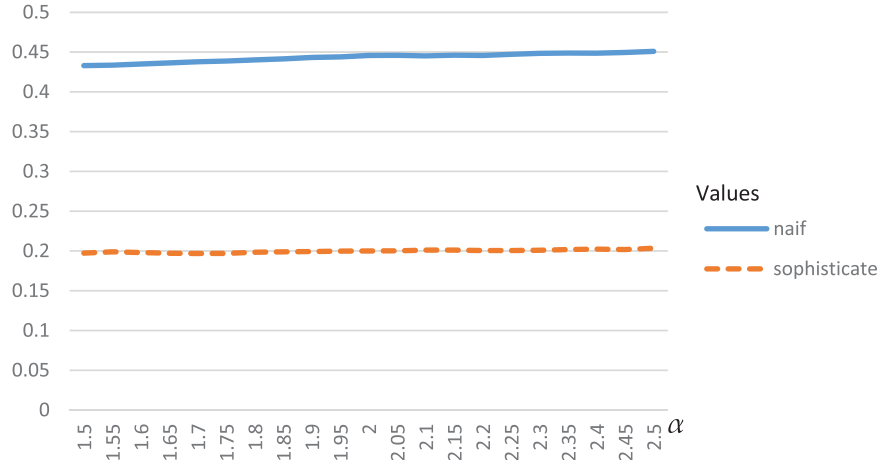


FIGURE 5 Percentage of profit loss by rejection as a function of α [Color figure can be viewed at wileyonlinelibrary.com]



preproperate project processing more than a naif. We observe that a naif only discounts his future values once when he makes a decision. However, a sophisticate applies compound discounting for revenues that occur later when he makes a decision. Thus, when completed projects can provide positive net profit, a naif is more likely to overestimate the future values of later projects and hence is more willing to delay a currently available project or even reject it.

4 | SEQUENCING DECISIONS

4.1 | Immediate costs

A TC's sequencing decision constitutes a classical scheduling problem. We consider the sequencing problem faced by a naif with immediate costs and revenues at project completion. Recall that a naif assumes he will be time consistent in all later decisions.

Consider the following algorithm for a naif. Let $F^{TC}(J, t)$ denote the maximum net profit a TC obtains by scheduling projects of set J , starting at time t . Recall that an optimal TC schedule characterized by $F^{TC}(J, t)$ may not process all the projects in J , since we allow project rejection.

Algorithm naive immediate costs for sequencing (NICS)

Given $p_1, \dots, p_n$, $v_1(\cdot), \dots, v_n(\cdot)$, $f_1(\cdot), \dots, f_n(\cdot)$, and $\beta_0, \beta_1, \dots, \beta_P$, where $P = \sum_{j=1}^n p_j$.

- Step 0. Set $t = 0$ and $J = \{1, \dots, n\}$.
- Step 1. For each project $j \in J$, compute $z(j) = -(1+r)^{-t}f_j(t+p_j) + \beta_t[(1+r)^{-(t+p_j)}v_j(t+p_j) + F^{TC}(J \setminus \{j\}, t+p_j)]$.
- Step 2. Let $j^* = \operatorname{argmax}_{j \in J} \{z(j)\}$. If $z(j^*) \geq 0$, then schedule project j^* from t to $t+p_{j^*}$; otherwise, reject all the projects in J and stop.
- Step 3. Set $t = t+p_{j^*}$ and $J = J \setminus \{j^*\}$, and if $J \neq \emptyset$ then go to Step 1.

Proposition 3. If a TC's sequencing problem can be solved optimally in $O(G)$ time, then Algorithm NICS solves the sequencing problem for a naif with immediate costs in $O(n^2G)$ time.

Proof. At any time given t , for any given set J of unprocessed projects, Algorithm NICS chooses the project to process next that maximizes a naif's perceived profit at time t . If the maximized profit at time t for projects in J is negative, then the naif cannot perceive any positive value at time t for

any subset of projects in J and rejects all the projects in J . Therefore, Algorithm NICS is optimal from the viewpoint of a naïf. Regarding the time complexity, Step 1 is applied n times, and each application calls an optimal algorithm with time complexity bounded by $O(G)$ for a TC no more than n times. Hence, the time complexity of Algorithm NICS is $O(n^2G)$. $\square$

We observe that an $O(G)$ time optimal algorithm in Proposition 3 may not necessarily be a polynomial time algorithm. However, there are cost functions for which the time-consistent sequencing problem is polynomially solvable, such as $f_j = C_j$ for the single processor considered here (Engels et al., 2003). Also, we observe that the project sequence identified by Algorithm NICS is not necessarily optimal for a TC. We now explore these two issues.

Example 3. The scheduling cost function is the weighted completion time, that is, $f_j(C_j) = w_j C_j$. The present bias coefficient is $\beta = 0.5$, and the discount rate is $r = 0$. There are $n = 2$ projects with $p_1 = 1$, $w_1 = 1$, $p_2 = 2$, and $w_2 = 3$. The project revenues v_1 and v_2 are any values that are sufficiently large such that rejection does not occur in an optimal decision.

To minimize the total weighted completion time, for a TC, the shortest weighted processing time first sequence, where projects are sequenced by the nondecreasing ratio of processing time to weight p_j/w_j , is optimal (Smith, 1956). For Example 3, the optimal sequence for a TC is $2 \rightarrow 1$, with a net profit of $v_1 + v_2 - 9$. However, for a naïf with immediate costs, at time zero, sequence $2 \rightarrow 1$ provides a net perceived profit of $0.5(v_1 + v_2) - 3(2) - 0.5(1)3 = 0.5(v_1 + v_2) - 7.5$, whereas $1 \rightarrow 2$ provides a greater net perceived profit of $0.5(v_1 + v_2) - 1(1) - 0.5(3)3 = 0.5(v_1 + v_2) - 5.5$ and hence is optimal.

In the problem considered below, the scheduling cost is the weighted number of late projects. For a TC, there exists a pseudopolynomial time algorithm (Lawler & Moore, 1969), but the problem is binary NP-hard (Karp, 1972).

Proposition 4. Consider a scheduling cost function is $f_j(C_j) = w_j U_j(C_j)$, where each project j has a time independent value v_j , a weight w_j , and a due date d_j , and $U_j(C_j) = 1$ if $C_j > d_j$ and $U_j(C_j) = 0$ if $C_j \leq d_j$. For this problem, there exists a sequence that is optimal for both a TC and a naïf.

Proof. Consider a schedule in which the on-time projects are sequenced by nondecreasing due dates, followed by the late projects with nondecreasing weights. This sequence is optimal for a TC (Lawler & Moore, 1969). Assume projects are sequentially indexed as $1, \dots, n$, where projects $1, \dots, i$ are on time and projects $i + 1, \dots, n$ are late.

To show the optimality of sequence $1, \dots, n$ for a naïf, supposing projects $1, \dots, j - 1$ have been processed with $C_{j-1} = t$ by the naïf, it suffices to show it is optimal for the naïf to process project j next. There are two cases to consider for the position of project j .

First, when $j \leq i$, processing project j next results in projects $j + 1, \dots, i$ being on time whereas projects $i + 1, \dots, n$ are late, and thus provides an intertemporal profit of $\beta_t(\sum_{k=j}^n v_k - \sum_{k=i+1}^n w_k)$ for projects $j, \dots, n$ to the naïf, where the present bias coefficient β_t applies following the definition of present bias as in (1), $\sum_{k=j}^n v_k$ is the total value of the on-time projects, and $\sum_{k=i+1}^n w_k$ is the total weight of the late projects. Because $\sum_{k=j}^n v_k$ is independent of sequencing decisions and $\sum_{k=i+1}^n w_k$ is minimized from the optimality of sequence $1, \dots, n$ for a TC, a naïf cannot improve such an intertemporal profit.

Second, when $j > i$, processing project j next earns an intertemporal profit of $w_j + \beta_t \sum_{k=j}^n (v_k - w_k)$ for projects $j, \dots, n$ to the naïf, where cost w_j is incurred immediately with no present bias coefficient, and $\beta_t \sum_{k=j}^n (v_k - w_k)$ is the total net profit from the projects $j, \dots, n$ each with a value and a tardy weight. The intertemporal profit $w_j + \beta_t \sum_{k=j}^n (v_k - w_k)$ cannot be improved due to the fact that the project values are time independent, and projects $j, \dots, n$ are in nondecreasing order of weights, which minimizes the total weight of the late projects. Therefore, sequence $1, \dots, n$ is optimal for both the TC and the naïf. $\square$

For a general scheduling cost function, the following algorithm optimizes the sequencing decisions of a sophisticate.

Algorithm sophisticated immediate costs for sequencing (SICS)

Input

Given $p_1, \dots, p_n$, $v_1(\cdot), \dots, v_n(\cdot)$, $f_1(\cdot), \dots, f_n(\cdot)$, and $\beta_0, \beta_1, \dots, \beta_P$, where $P = \sum_{j=1}^n p_j$. Let $N = \{1, \dots, n\}$.

Value function

$F_c^S(J, t)$ = the maximum net profit perceived at time t by a sophisticate of any schedule of projects with immediate costs in set J , in which the starting time of any project in J is no earlier than t .

Boundary condition

$$F_c^S(\emptyset, t) = \begin{cases} 0, & \text{for } 0 \leq t \leq P, \\ -\infty, & \text{for } t > P. \end{cases} \quad (18)$$

Optimal solution value

$F_c^S(N, 0)$.

Recurrence relation

$$F_c^S(J, t) = \max \begin{cases} \max_{j \in J} \{ -(1+r)^{-t} f_j(t + p_j) \\ + \beta_t [(1+r)^{-(t+p_j)} v_j(t + p_j) \\ + F_c^S(J \setminus \{j\}, t + p_j)] \}, \\ \max_{j \in J} \{ F_c^S(J \setminus \{j\}, t) \}. \end{cases} \quad (19)$$

In the recurrence relation, in the first alternative, a project $j \in J$ is selected to start processing at time t to maximize the intertemporal profit of the sophisticate at time t . The second alternative rejects a project $j \in J$ and maximizes the

intertemporal profit of the sophisticate at time t for projects $J \setminus \{j\}$.

The optimality of Algorithm SICS follows from its implicit enumeration in the recurrence relation. In the value function, there are up to 2^n different sets J and up to P different values of t . The recurrence relation selects j from J , which requires up to n comparisons. Therefore, the overall time complexity of the algorithm is $O(n2^n P)$.

4.2 | Immediate revenues

We consider the sequencing problem, for a naïf, with immediate revenues and costs at project completion. Let $F^{TC}(J, t)$ denote the maximum net profit a TC earns by scheduling projects of set J , starting at time t . We propose the following algorithm.

Algorithm naïf immediate revenues for sequencing (NIRS)

Step 0. Set $t = 0$ and $J = \{1, \dots, n\}$.

Step 1. For each project $j \in J$, compute $z(j) = (1 + r)^{-t} v_j(t + p_j) + \beta_t [-(1 + r)^{-(t+p_j)} f_j(t + p_j) + F^{TC}(J \setminus \{j\}, t + p_j)]$.

Step 2. Let $j^* = \arg\max_{j \in J} z(j)$. If $z(j^*) \geq 0$, then process project j^* from t to $t + p_{j^*}$; otherwise, reject projects in J and stop.

Step 3. Set $t = t + p_{j^*}$ and $J = J \setminus \{j^*\}$, and if $J \neq \emptyset$ then go to Step 1.

Proposition 5. *If a TC's sequencing problem can be solved optimally in $O(G)$ time, then from the viewpoint of a naïf, Algorithm NIRS solves the sequencing problem with immediate revenues in $O(n^2 G)$ time.*

The proof of Proposition 5 is similar to that of Proposition 3 and is therefore omitted.

With immediate revenues and project completion time cost, the shortest processing time first (SPT) rule which is optimal for a TC is not necessarily optimal for a naïf. This is because, with immediate revenues, project revenue strongly influences the total net profit of a schedule. Consider the following example.

Example 4. The scheduling cost function is the project completion time, that is, $f_j(C_j) = C_j$. The time independent present bias coefficient is $\beta = 0.5$, and the discount rate is $r = 0$. There are $n = 2$ projects with $p_1 = 1$, $v_1 = 4$, $p_2 = 2$, and $v_2 = 8$.

For Example 4, the SPT sequence $1 \rightarrow 2$ provides a net perceived profit of $4 + 0.5(8 - 1 - 3) = 6$, while the reversed SPT sequence $2 \rightarrow 1$ delivers a perceived net profit of $8 + 0.5(4 - 2 - 3) = 7.5$. Processing only project 1 generates a perceived net profit of $4 + 0.5(-1) = 3.5$, and processing only project 2 generates $8 + 0.5(-2) = 7$. Hence, a naïf processes project 2 first and project 1 second.

Similar to the case with immediate costs, we present the following algorithm that optimizes the sequencing decision of a sophisticate with immediate revenues.

Algorithm sophisticated immediate revenues for sequencing

Value function

$F_r^S(J, t)$ = the maximum total net profit perceived at time t by a sophisticate of any schedule of projects with immediate revenues in set J , in which the starting time of any project in J is no earlier than t .

Boundary condition

$$F_r^S(\emptyset, t) = \begin{cases} 0, & \text{for } 0 \leq t \leq P, \\ -\infty, & \text{for } t > P. \end{cases} \quad (20)$$

Recurrence relation

$$F_r^S(J, t) = \max \begin{cases} \max_{j \in J} \{ (1 + r)^{-t} v_j(t + p_j) \\ + \beta_t [-(1 + r)^{-(t+p_j)} f_j(t + p_j) \\ + F_r^S(J \setminus \{j\}, t + p_j)] \}, \\ \max_{j \in J} \{ F_r^S(J \setminus \{j\}, t) \}. \end{cases} \quad (21)$$

4.3 | Managerial insights for sequencing decisions

As in Section 3.3, we first investigate with scheduling cost defined as project completion time, that is, $f_j(C_j) = C_j$, how present bias affects a naïf's and a sophisticate's sequencing decisions.

Theorem 2. *If $r = 0$, $f_j(C_j) = C_j$, and project values v_j are sufficiently large to prevent rejection, with time independent present bias coefficient β , then*

- (i) *with immediate costs, a naïf processes his projects by nondecreasing order of project processing time, as does a TC,*
- (ii) *with immediate costs, a sophisticate processes his projects by nondecreasing order of $p_j - \beta(1 - \beta)v_j$,*
- (iii) *with immediate revenues, both a naïf and a sophisticate process their projects by nondecreasing order of $\beta p_j - (1 - \beta)v_j$.*

Proof. For part (i), we first note that the SPT rule is optimal for a TC, that is, projects are sequenced by nondecreasing processing times. To see that the SPT rule is also optimal for a naïf with immediate costs, consider an interchange argument with two projects i and j , where $p_i \leq p_j$, that are processed consecutively. Perceived at any starting time by a naïf, sequence $i \rightarrow j$ incurs a cost of $p_i + \beta(p_i + p_j)$, and sequence $j \rightarrow i$ incurs a cost of $p_j + \beta(p_j + p_i)$. Both sequences have a perceived revenue of $\beta(v_i + v_j)$. Therefore, sequence $i \rightarrow j$ weakly dominates sequence $j \rightarrow i$ when $p_i \leq p_j$.

For part (ii), consider a sophisticate's sequencing decision for the same problem. Now consider a pairwise interchange argument involving projects i and j . Consider the following alternative decisions for a sophisticate. The following expressions are written for profit rather than cost, where $F_c^S(J, t)$ denotes the maximum net profit for a sophisticate from scheduling projects of set J with immediate costs, starting at time t .

$$i \rightarrow j: -f_i(t + p_i) + \beta v_i - \beta f_j(t + p_i + p_j) + \beta^2 v_j + \beta^2 F_c^S(J \setminus \{i, j\}, t + p_i + p_j).$$

$$j \rightarrow i: -f_j(t + p_j) + \beta v_j - \beta f_i(t + p_j + p_i) + \beta^2 v_i + \beta^2 F_c^S(J \setminus \{i, j\}, t + p_j + p_i).$$

Comparing these expressions for profit, we have the difference in profit $\Delta = f_j(t + p_j) - f_i(t + p_i) + \beta f_i(t + p_j + p_i) - \beta f_j(t + p_i + p_j) + \beta(1 - \beta)(v_i - v_j)$.

With $f_j(C_j) = C_j$, we have $\Delta = (p_j - p_i) + \beta(1 - \beta)(v_i - v_j) = (p_j - \beta(1 - \beta)v_j) - (p_i - \beta(1 - \beta)v_i)$. This expression provides a universal, that is, transitive, sequencing rule that defines an optimal order of the projects: nondecreasing order of $p_j - \beta(1 - \beta)v_j$, which is the β -weighted difference between processing time and project revenue.

For part (iii), to find an optimal sequence for a naif, consider a pairwise interchange argument involving projects i and j , with the following expressions written for profit:

$$i \rightarrow j: v_i - \beta f_i(t + p_i) + \beta v_j - \beta f_j(t + p_i + p_j) + \beta F^{TC}(J \setminus \{i, j\}, t + p_i + p_j).$$

$$j \rightarrow i: v_j - \beta f_j(t + p_j) + \beta v_i - \beta f_i(t + p_j + p_i) + \beta F^{TC}(J \setminus \{i, j\}, t + p_j + p_i).$$

Comparing these expressions for profit, we have $\Delta = (v_i - v_j)(1 - \beta) + \beta[f_j(t + p_j) - f_i(t + p_i)] - \beta[f_j(t + p_i + p_j) - f_i(t + p_j + p_i)]$. With project cost defined as project completion time and immediate project revenues, we have $\Delta = (v_i - v_j)(1 - \beta) + \beta(p_j - p_i) = (\beta p_j - (1 - \beta)v_j) - (\beta p_i - (1 - \beta)v_i)$. This expression provides a universal sequencing rule that defines an optimal order of the projects: nondecreasing order of $\beta p_j - (1 - \beta)v_j$, which is the β -weighted difference between processing time and project revenue.

Finally, consider a sophisticate's sequencing decision for the same problem. Let $F_r^S(J, t)$ denote the maximum net profit for a sophisticate from scheduling projects of set J with immediate revenues, starting at time t . Consider a pairwise interchange argument involving projects i and j , with the following expressions written for profit:

$$i \rightarrow j: v_i - \beta f_i(t + p_i) + \beta v_j - \beta^2 f_j(t + p_i + p_j) + \beta^2 F_r^S(J \setminus \{i, j\}, t + p_i + p_j).$$

$$j \rightarrow i: v_j - \beta f_j(t + p_j) + \beta v_i - \beta^2 f_i(t + p_j + p_i) + \beta^2 F_r^S(J \setminus \{i, j\}, t + p_j + p_i).$$

Comparing these expressions for profit, we have $\Delta = (v_i - v_j)(1 - \beta) + \beta[f_j(t + p_j) - f_i(t + p_i)] - \beta^2[f_j(t + p_i + p_j) - f_i(t + p_j + p_i)]$. With project cost defined as project completion time and immediate project revenues, we have $\Delta = (v_i - v_j)(1 - \beta) + \beta(p_j - p_i) = (\beta p_j - (1 - \beta)v_j) - (\beta p_i - (1 - \beta)v_i)$, the same as for a naif. $\square$

We note that in part (ii) of Theorem 2, for a sophisticate with $f_j(C_j) = C_j$, the SPT rule is not necessarily optimal. Although shorter processing time promotes earlier processing, a project with a longer processing time may need to be processed earlier when its revenue is sufficiently large. This result is intuitive since, with future costs and revenues discounted by the present bias coefficient β , the decision-maker prefers to prioritize projects with shorter processing time and greater revenue. Similarly, we can show that for the problem $f_j(C_j) = w_j U_j(C_j)$ discussed in Proposition 4, an optimal sequence found for a TC is not necessarily optimal for a sophisticate.

In part (iii) of Theorem 2, for the sequencing problem with immediate revenues and cost function $f_j(C_j) = C_j$, a naif and a sophisticate have the same optimal decision. Compared with immediate costs instead of immediate revenues, for a sophisticate, the optimal sequence depends more strongly on the project revenue. That is, the weight of project revenue relative to processing time increases from $\beta(1 - \beta)$ to $(1 - \beta)/\beta$. Intuitively, this is because, with immediate revenues and present bias, a project's revenue becomes more important in determining the decision-maker's intertemporal profit at any given time.

Overall, Theorem 2 shows that the design of projects with immediate costs by project owners can help prevent biased decisions for the sequencing problem.

Next, we use a numerical experiment to study how present bias can lead to inefficient sequencing decisions. The experimental design is similar to that in Section 3.3. To focus on sequencing in a given time window in length equal to the total processing time of the given projects, we do not allow rejection. Note that for the sequencing problem with $f_j(C_j) = C_j$, in an optimal schedule, projects are processed in SPT order, where an individual project may incur a negative profit. For the three parameters we control, we let $r = 0$, $\beta \in \{0.4, 0.45, \dots, 0.95\}$, and $\alpha \in \{0.05, 0.1, \dots, 1\}$. We use smaller values of α to obtain similar profit margins to those in the timing problem since the sequencing problem provides a larger feasible region. Each figure is based on results from all the instances.

Figure 6 shows that the profit loss due to present bias is decreasing with β , there is a greater loss with immediate revenues, and there is a decreasing gap between the two mechanisms as β increases. Figure 7 depicts the same effects of α as in Figure 2.

We next discuss the managerial insights obtained from our study of sequencing decisions. Recall that the preference is always for an economically efficient time-consistent decision. First, from a computational perspective, a TC's sequencing decision is the easiest, while a sophisticate's decision is the hardest. This observation has a favorable implication; since a sequencing decision is harder to make using a naive or sophisticated strategy, a naif or sophisticate may use a time-consistent decision for ease of decision-making.

Second, for immediate costs, with certain scheduling cost functions, for example, project completion time cost, that is,

FIGURE 6 Percentage of profit loss in sequencing as a function of β [Color figure can be viewed at wileyonlinelibrary.com]

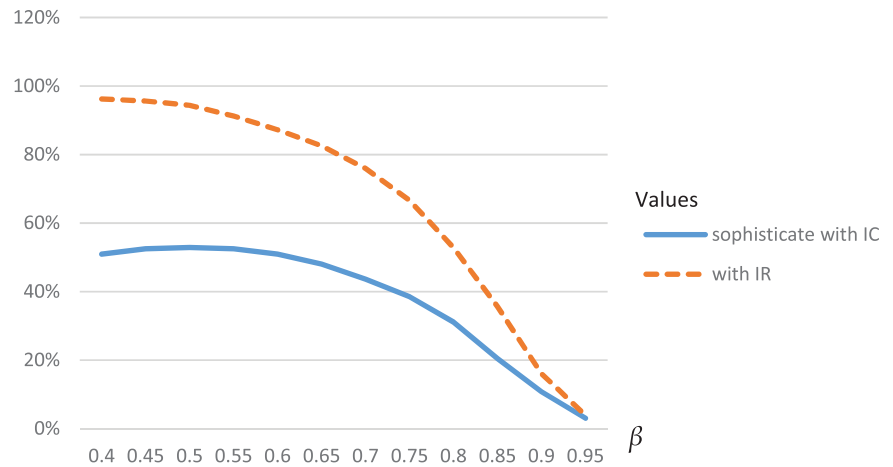
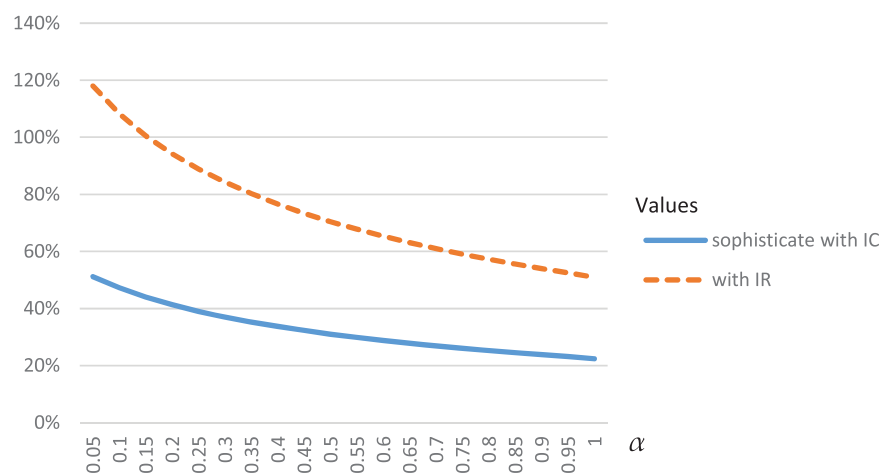


FIGURE 7 Percentage of profit loss in sequencing as a function of α [Color figure can be viewed at wileyonlinelibrary.com]



$f_j(C_j) = C_j$, and weighted number of late projects, that is, $f_j(C_j) = w_j U_j$ as in Proposition 4, a TC and a naif have a common optimal sequence. This implies that present bias may not necessarily lead to an inefficient decision.

Third, an optimal sequence under the naive or sophisticated strategy is closer to an optimal sequence as found by a time-consistent strategy under immediate costs than under immediate revenues when sequencing decisions influence project costs more than project revenues. Intuitively, it is better to have either cost or revenue, whichever is more affected by sequencing decisions, at the beginning of a project to promote more efficient sequencing by project companies.

Fourth, we observe a tendency to process projects with more significant revenue and negligible cost earlier under biased decisions than under time-consistent decisions. This is because present bias often prioritizes projects with a greater value-to-cost ratio. To mitigate the inefficiency that results from this tendency, project owners may assign costs and revenues more equally among projects or assign costs and revenues proportionately for projects during project contracting.

5 | CONCLUDING REMARKS

This work is among the first to study the effect of the well-documented phenomenon of present bias within a scheduling context. Within a simple scheduling system, we use examples to illustrate time-inconsistent decision-making and describe algorithms to optimize revenue less cost for both naive and sophisticated time-inconsistent decision-makers. This enables us to develop insights into the relative performance of time-consistent, naive, and sophisticated decision-makers, and how to mitigate the effects of present bias. These insights are validated both by theoretical results and computational studies.

This work studies how present bias, as modeled by a present bias coefficient and a level of self-awareness of this bias, affects decisions in the scheduling of projects, as well as how to mitigate biases in such decisions. First, we study timing decisions for a fixed sequence of potential projects. Here, we find that it is helpful for a project owner, in anticipation of present bias by a project company, to offer a contract with immediate revenues, since doing so may avoid delay or rejection of an urgent project. Further, to encourage postponement

of a project strategically, a project owner may offer one with immediate costs. For sequencing decisions between projects, it is beneficial for a project owner to have either cost or revenue, whichever is mainly affected by sequencing decisions, scheduled immediately at the beginning of a project, in order to promote more efficient sequencing by a project company.

For future research, an extension of our work is to consider scheduling decisions under present bias in different bias and awareness formats other than rate and naïf/sophisticate awareness. We also recommend further comparisons of time-consistent and time-inconsistent decision-making for operational decisions. The extensive classical scheduling literature (Demeulemeester & Herroelen, 2002; Pinedo, 2016) describes numerous directions by which to explore this comparison. In addition, models of simple inventory trade-offs that require decision-making over time under present bias, along with measures to mitigate its effects in that context, define interesting problems for study.

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